

AI 2002

Artificial Intelligence

Lecture 13

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FAST NUCES CFD

$$\alpha \models \beta \text{ if and only if } M(\alpha) \subseteq M(\beta)$$

Logic - Valid

▶ A sentence is **valid** if it is *true* in all models (interpretations).

▶ **Valid Sentences:**

true, $\neg$ false, $P \vee \neg P$

▶ Valid sentences are also known as **tautologies**—they are *necessarily true*.

▶ From the definition of entailment, we can derive the **deduction theorem**, for example

For any sentences α and β , $\alpha \models \beta$ if and only if the sentence $(\alpha \Rightarrow \beta)$ is valid.

Logic: satisfiability

- A sentence is **satisfiable** if it is *true* in, or satisfied by, some model.
- A sentence is **satisfiable** if and only if it's *true* in at least one interpretation.
- Satisfiable Sentence:
 - true, $P, \neg P$
- A sentence is **unsatisfiable** if and only if its truth value is *false* in all interpretations.
 - $\neg$ true, false, $P \wedge \neg P$

Examples:

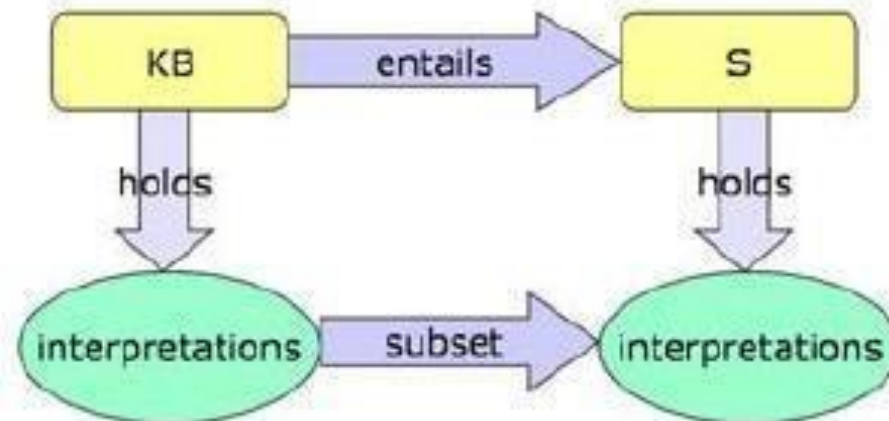
Sentences	Valid/ Satisfiability?	Interpretation
Smoke $\rightarrow$ Smoke	Valid	
Smoke $\vee$ $\neg$ Smoke	Valid	
Smoke $\rightarrow$ Fire	Satisfiable	Smoke = t Fire = f , "smoke implies fire"
$(s \rightarrow f) \rightarrow (\neg s \rightarrow \neg f)$	Satisfiable	$s = \mathbf{f}, f = \mathbf{t}$ $(s \text{ e } f) = \mathbf{t}, (\neg s \text{ e } \neg f) = \mathbf{f}$ "Smoke implies fire implies not smoke implies not fire."
$(s \rightarrow f) \rightarrow (\neg f \rightarrow \neg s)$	Valid	Contrapositive "smoke implies fire implies not fire implies not smoke"

Logic: satisfiability

- The problem of satisfiability of a sentence is very **related** to the **constraint satisfaction problem**, that has to **find a legal assignment** that can satisfy the constraint,
- Similarly, we have to ***find interpretation*** *in such as way that sentence holds that interpretation.*
- Can be solved the problem using the **brute-force method** of **enumerating all possible interpretations.**

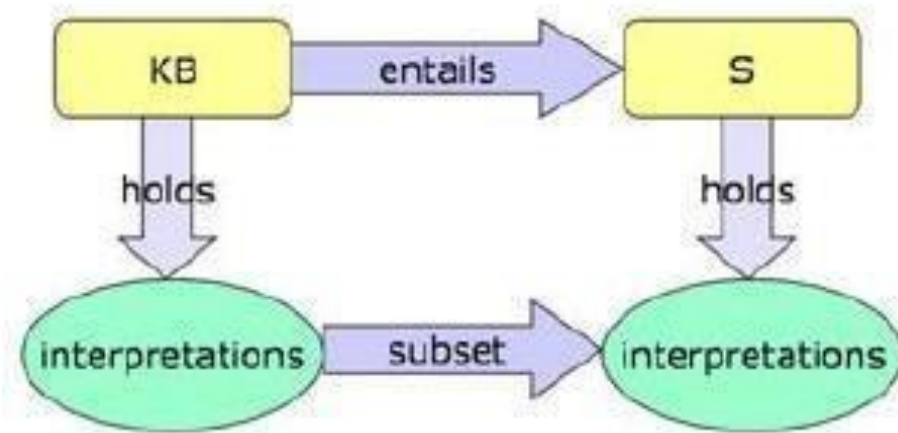
Entailment

- Entailment that signifies a relationship between a knowledge base and another sentence.
- If every interpretation that satisfies the *KB* also satisfies the conclusion (sentence), we'll say that the
- *KB* "entails" the conclusion (sentence).



Entailment

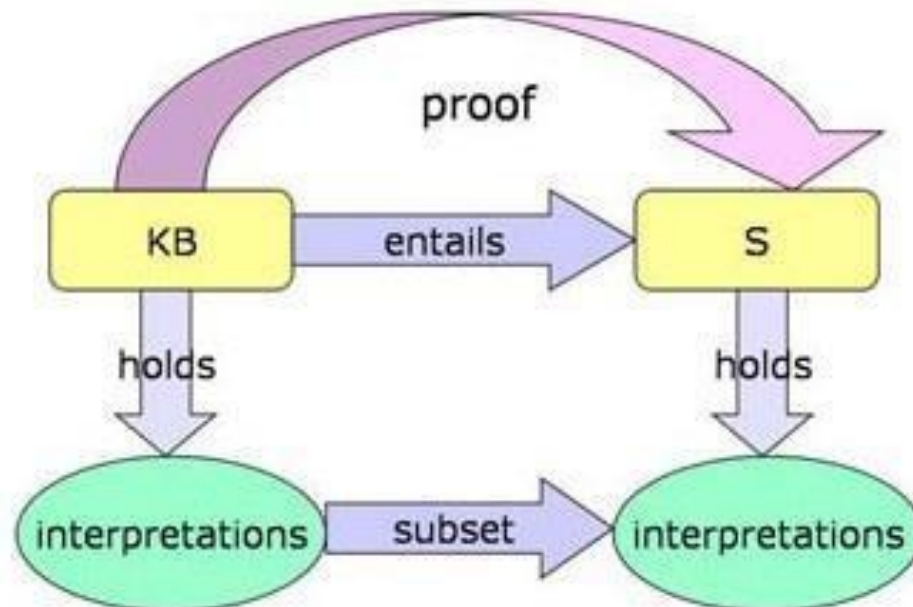
- We have to enumerate all interpretations
- Those interpretations are selected where all elements of ***KB*** are true.
- Check if the sentence ***S*** is true for all those interpretations.



Too many interpretations in general!

Entailment and proof

- Proof is a way to test whether a *KB* entails a sentence without enumerating too many interpretations.
- Proof is basically a sequence of sentences.
- Proof is a chain of conclusions that leads to the desired goal.



Inference

- If an inference algorithm i can derive α from KB , we write

$$KB \vdash_i \alpha$$

- which is pronounced “ α is derived from KB by i ” or “ i derives α from KB .”

Inference Rules

- Some of Inference rules can be described as,

- **Modus Ponens**

- ▶ If the sentences $\alpha \rightarrow \beta$ and α are known to be true, then **modus ponens** lets us infer β .

$$\frac{\alpha \rightarrow \beta \quad \alpha}{\beta}$$

- **Modus Tollens**

- ▶ Under the inference rule **modus tollens**, if $\alpha \rightarrow \beta$ is known to be true and $\neg \beta$ is known to be false, we can infer $\neg \alpha$.

$$\frac{\alpha \rightarrow \beta \quad \neg \beta}{\neg \alpha}$$

Inference Rules

- **And Elimination:**

- ▶ And **elimination** allows us to infer the truth of either of the conjuncts from the truth of a conjunctive sentence. For instance, $\alpha \wedge \beta$ lets us conclude α and β are true.

$$\frac{\alpha \wedge \beta}{\alpha}$$

- **And Introduction:**

- ▶ And introduction lets us infer the truth of a conjunction from the truth of its conjuncts. For instance, if α and β are true, then $\alpha \wedge \beta$ is true.

$$\frac{\alpha \quad \beta}{\alpha \wedge \beta}$$

Inference rule and proof

- Inference rules can be applied to derive a proof.

$$\frac{\alpha \rightarrow \beta \quad \alpha}{\beta}$$

Modus
ponens

$$\frac{\alpha \rightarrow \beta \quad \neg \beta}{\neg \alpha}$$

Modus
tolens

$$\frac{\alpha \quad \beta}{\alpha \wedge \beta}$$

And-
introduction

$$\frac{\alpha \wedge \beta}{\alpha}$$

And-
elimination

Example:

- Consider there are 3 sentences in our knowledge base, $P \wedge Q$, $P \rightarrow R$ and $(Q \wedge R) \rightarrow S$. **We have to prove S.**

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Step	Formula	Derivations
1	$P \wedge Q$	Given
2	$P \rightarrow R$	Given
3	$(Q \wedge R) \rightarrow S$	Given
4	P	1 And-Elimination
5	R	2,4 Modes Ponens
6	Q	1 And-Elimination
7	$Q \wedge R$	6,5 And-Introduction
8	S	3,7 Modes Ponens

Example 2:

- Consider there are 5 sentences in our knowledge base as given below.
 - We have to prove R**
- | Step | Formula | Derivations |
|------|---------|-------------|
| | | |

[illegible]

Example 2:

- ▶ Consider there are 5 sentences in our knowledge base as given below.

- ▶ **We have to prove R**

Step	Formula	Derivations
1	$P \rightarrow Z$	Given
2	$(P \wedge R) \rightarrow S$	Given
3	$(Q \wedge Z) \rightarrow R$	Given
4	$Z \rightarrow Q$	Given
5	$P \wedge S$	Given
6	P	5 And-Elimination
7	Z	6,1 Modes Ponens
8	Q	4,7 Modes Ponens
9	$Q \wedge Z$	7,8 And introduction
10	R	3,9 Modes Ponens

Propositional Resolution

- ▶ Resolution is one method for **automated theorem proving**.
- ▶ It helps logical agents **to reason** about the world.
- ▶ It helps logical agents **to prove new theorems**, and therefore helps them **to add to their knowledge**.
- ▶ Resolution requires all sentences to be first written in a special form - **conjunctive normal form**.

Propositional Resolution

- ▶ Input a knowledge base and an expression.
- ▶ It **negates the expression**, adds that to the knowledge base, and finds a **contradiction** if one exists
- ▶ If it **finds** a contradiction, then the negated statement is **false**.
- ▶ Therefore, the original statement **must be true**.

Conjunctive Normal Form

- ▶ A series of “conjunctions” (clauses joined together by “ $\wedge$ ”).

$$(A \vee B \vee \neg C) \wedge (B \vee D) \wedge (\neg A) \wedge (B \vee C)$$

clauses

- ▶ A, B, $\neg C$ are **literals**.
- ▶ Inside the brackets, we only have “ $\vee$ (**OR**)”, symbols.

There must be no “implies” ($\rightarrow$) symbols anywhere.

Conjunctive Normal Form

- ▶ **Resolution algorithm** ‘resolves’ clauses
- ▶ Each pair of clauses that contains **complementary literals** is resolved.
Complementary literals have the property that one negates the other
 - $P, \neg P$
 - the clause $B_{1,1} \vee \neg B_{1,1} \vee P_{1,2}$ is equivalent to $True \vee P_{1,2}$ which is equivalent to $True$.
Deducing that $True$ is true is not helpful.
- ▶ Therefore, *any clause in which two complementary literals appear can be discarded.*
- ▶ Each clause is a requirement that must be satisfied.
- ▶ It can be satisfied in multiple ways

Converting in CNF

1. Biconditional to CNF

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

2. Implication to CNF

$$(\alpha \Rightarrow \beta) \equiv (\neg \alpha \vee \beta) \quad \text{implication elimination}$$

Converting in CNF

3. De Morgan Law

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{De Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{De Morgan}$$

4. Distributivity of $\wedge$ and $\vee$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

Example: Converting to CNF

$$(A \vee B) \rightarrow (C \rightarrow D)$$

1. Eliminate arrows

$$\neg(A \vee B) \vee (\neg C \vee D)$$

2. Drive in negations

$$(\neg A \wedge \neg B) \vee (\neg C \vee D)$$

3. Distribute

$$(\neg A \vee \neg C \vee D) \wedge (\neg B \vee \neg C \vee D)$$

Resolution Rule

Resolution rule:

$$\alpha \vee \beta$$

$$\neg\beta \vee \gamma$$

$$\alpha \vee \gamma$$

Resolution Theorem

1. Convert all sentences into CNF
2. Negate the desired conclusion (converted to CNF)
3. Apply resolution rule until either
 - Derive false (contradiction)
 - The conclusion follows from the axioms
 - Can not apply anymore,
 - The conclusion can not be proved from axioms
- 4. Propositional Resolution is sound and complete.**

Example:

Consider we are given "P or Q", "P implies R" and "Q implies R". We would like to conclude R from these three axioms

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

Solution:

Step 1: Negated the conclusion

Steps	Formula	Derivations
1	$P \vee Q$	Given
2	$P \rightarrow R$	Given
3	$Q \rightarrow R$	Given
4	$\sim R$	Conclusion

Solution:

- Step 1: Negated the conclusion
- Step 2 : Eliminate the Implication

$$P \rightarrow R = \sim P \vee R$$

$$Q \rightarrow R = \sim Q \vee R$$

Steps	Formula	Derivations
1	$P \vee Q$	Given
2	$\sim P \vee R$	Given
3	$\sim Q \vee R$	Given
4	$\sim R$	Conclusion

Solution:

Steps	Formula	Derivations
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	Negated Conclusion
5	$Q \vee R$	1,2 Resolution Rule
6	$\neg P$	2,4
7	$\neg Q$	3,4
8	R	5,7
9	.	4,8

Example 2:

- Consider we are given following table. We would like to conclude R from these three axioms

1	$(P \rightarrow Q) \rightarrow Q$
2	$(P \rightarrow P) \rightarrow R$
3	$(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$

Step 1:

1	$(P \rightarrow Q) \rightarrow Q$
2	$(P \rightarrow P) \rightarrow R$
3	$(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$

Step	Formula & Derivations
1	$(P \rightarrow Q) \rightarrow Q$
2	$(\neg P \vee Q) \rightarrow Q$
3	$\neg(\neg P \vee Q) \vee Q$
4	$(P \wedge \neg Q) \vee Q$
5	$(P \vee Q) \wedge (\neg Q \vee Q)$
6	$(P \vee Q)$

Step 2:

1	$(P \rightarrow Q) \rightarrow Q$
2	$(P \rightarrow P) \rightarrow R$
3	$(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$

Step	Formula & Derivations
1	$(P \rightarrow P) \rightarrow R$
2	$(\neg P \vee P) \rightarrow R$
3	$\neg(\neg P \vee P) \vee R$
4	$(P \wedge \neg P) \vee R$
5	$(P \vee R) \wedge (\neg P \vee R)$

Step 3:

1	$(P \rightarrow Q) \rightarrow Q$
2	$(P \rightarrow P) \rightarrow R$
3	$(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$

Step	Formula & Derivations
1	$(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$
2	$(\neg R \vee S) \rightarrow \neg(\neg S \vee Q)$
3	$\neg(\neg R \vee S) \vee \neg(\neg S \vee Q)$
4	$(R \wedge \neg S) \vee (S \wedge \neg Q)$
5	$(R \vee S) \wedge (R \vee \neg Q) \wedge (\neg S \vee S) \wedge (\neg S \vee \neg Q)$
6	$(R \vee S) \wedge (R \vee \neg Q) \wedge (\neg S \vee \neg Q)$

Solution:

1	$(P \rightarrow Q) \rightarrow Q$
2	$(P \rightarrow P) \rightarrow R$
3	$(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$

Step	Formula	Derivations
1	$(P \vee Q)$	
2	$(P \vee R)$	
3	$(\neg P \vee R)$	
4	$(R \vee S)$	
5	$(R \vee \neg Q)$	
6	$(\neg S \vee \neg Q)$	
7	$\neg R$	Negation

Solution

1	$(P \rightarrow Q) \rightarrow Q$
2	$(P \rightarrow P) \rightarrow R$
3	$(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$

Step	Formula	Derivations
1	$(P \vee Q)$	
2	$(P \vee R)$	
3	$(\neg P \vee R)$	
4	$(R \vee S)$	
5	$(R \vee \neg Q)$	
6	$(\neg S \vee \neg Q)$	
7	$\neg R$	Negation
8	S	4,7
9	$\neg Q$	6,8
10	P	1,9
11	R	3,10
12	.	7,11